

Math Systems Biology
Slides to accompany lecture notes:
Difference Equations

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Introduction

material based on part of Chapter 1 of
Mathematical Models in Biology: An Introduction, 2004
by Allman and Rhodes

see book for much more material!

with differential eqs, behaviors interesting only in higher dimensions
but with difference eqs, behaviors such as periodic orbits and chaos
can appear in models with just one variable

also: easier to understand
& provides an easy introduction to ideas about dynamics

Difference equations

time variable discrete, $t = 0, 1, 2, \dots$

implicitly assumed fixed a unit of time measurement

e.g. t may be measured in generations in genetics,
or hours, days, years if population size for cells, flies, humans

$P(t)$ = size of population (or another quantity) at time t

sometimes more convenient to write " P_t " instead of $P(t)$

difference equation: a rule that tells how
population at time $t + 1$ depends on population at time t :

$$P_{t+1} = F(P_t)$$

usually too hard to find closed-form solutions

numerical experiments can help tremendously
if coupled with qualitative understanding

Iterations and P^+ and Δ formalisms

sometimes more convenient (and better analogy to differential eqs) to specify *change* in population

$$\Delta P_t := P_{t+1} - P_t$$

obviously can compute P_{t+1} from P_t by adding back change:

$$P_{t+1} = P_t + \Delta P_t$$

so may specify either ΔP_t or P_{t+1} as functions of P_t
convention: instead of $P_{t+1} = F(P_t)$, we sometimes write

$$P^+ = F(P)$$

(read superscript “+” as “step the index up by one”)
and instead of ΔP_t we write ΔP , dropping t

Linear equations

exponential growth (Malthus, late 18th century)

suppose at each time t , population *change* is:

1. add multiple f (“fecundity” due to births) to population and
2. subtract a fraction d (death rate) from population

i.e.

$$\Delta P_t = fP_t - dP_t = (f - d)P_t$$

or equivalently:

$$P^+ = P + \Delta P = P + (f - d)P = (1 + f - d)P = \lambda P,$$

where $\lambda := 1 + f - d$ growth rate of the population

Simple Example

$f = 0.1$, $d = 0.02$, so that $\lambda = 1 + f - d = 1.08$
if we specify an “initial condition” such as $P_0 = 100$,
we may recursively compute:

$$P^+ = 1.08P, \quad P_0 = 100$$

by repeatedly multiplying by 1.08. We get

Day	Population
0	100
1	$(1.08)100 = 108$
2	$(1.08)^2 100 = 116.64$
3	$(1.08)^3 100 \approx 125.971$
4	$(1.08)^4 100 \approx 136.0489$

and, obviously, we know the solution for all t in closed form:
 $P = 100(1.08)^t$

Sample “word problem”

study insect species for which, in each generation and assume:
each female lays 150 eggs (in the average)

when eggs hatch, only 2% survive to become adult females
(remaining eggs are assumed to not hatch or to be males)

time measured in generations,

all adults die before next generation size computed

so death rate is $d = 1$ (everyone dies) and the effective fecundity is

$$f = .02(150) = 3$$

which leads to the following equation for the female population:

$$P^+ = (1 + 3 - 1)P = 3P$$

(ignored males in order to simplify the model;
assume enough males available)

Non-integers?

What is the meaning of non-integer populations?

depends on the units in which P is being specified

e.g. if P_t measured in millions of individuals (cells, people)

fractions have obvious meaning: 136.0489 is 1,360,489 individuals

or may be measuring populations by weight, such as in tons

if dealing with a harvest of a crop or fish

anyway, models are ultimately simplifications of reality,

so “136.0489” is just “approximately 136”

Per-capita growth rate

exponential growth model not realistic but, in practice,
birth and death rates are dependent on available resources
(such as food, water, space)

so, at least indirectly, on population size

so reasonable to modify linear growth model by assuming
growth rate decreases as the population size increases
(density-dependent growth)

consider *per-capita* growth rate over a single time step

i.e. change in population per individual $\Delta P_t / P_t$

in simple exponential growth model, $\Delta P_t / P_t \equiv \lambda$ constant

Carrying capacity

small populations: sufficient resources, per capita growth rate $r > 0$

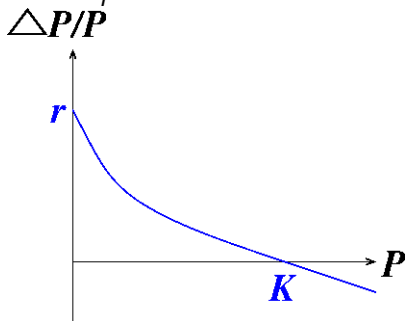
large populations: slower per-capita growth (competition)

and if populations $> K$ ("carrying capacity" of environment)

then per-capita growth rate < 0

(starvation, incapability to find mates, etc)

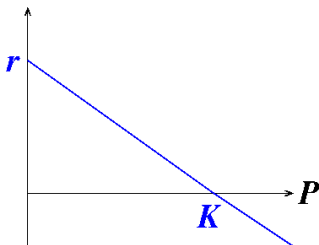
thus want function s.t. $\frac{\Delta P}{P}$ decreases with increasing P , eventually < 0



Simplest per-capita growth: logistic model

simplest such function is linear

$\Delta P/P$



$$\frac{\Delta P}{P} = r \left(1 - \frac{P}{K}\right).$$

i.e. $\Delta P_t = rP_t \left(1 - \frac{P_t}{K}\right)$, or since $P_{t+1} = P_t + \Delta P_t$:

$$P^+ = P \left(1 + r \left(1 - \frac{P}{K}\right)\right)$$

note if population small ($P \ll K$), $P/K \approx 0$ so:

$$P^+ \approx \lambda P, \quad \text{where } \lambda = 1 + r$$

i.e. when population $\ll$ carrying capacity, same as simple model

Simple behavior (for some parameters!)

no simple expression for the iterates, comparable to " $\lambda^t P_0$ "

explore numerically to have some feel for solutions

amazingly, this model can lead to very interesting and unexpected behaviors!

but for some parameters, behavior not surprising, e.g. iterate

$$P^+ = P(1 + .7(1 - P/10)), \quad P_0 = 0.4346$$

to obtain:

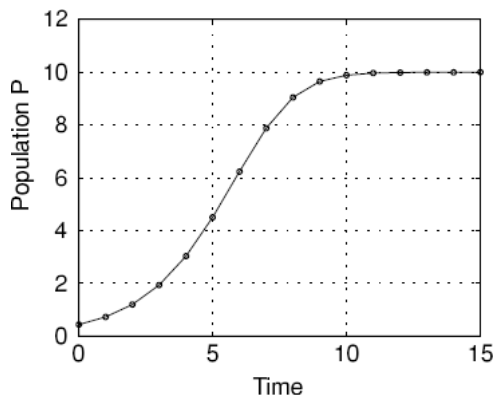
$$P_1 = .7256, P_2 = 1.1967, P_3 = 1.9341,$$

$$P_4 = 3.0262, P_5 = 4.5034, P_6 = 6.2362,$$

$$P_7 = 7.8792, P_8 = 9.0489, P_9 = 9.6514,$$

$$P_{10} = 9.8869, P_{11} = 9.9652, P_{12} = 9.9895, \dots$$

Simple behavior for some parameters, ctd'



(only values at integer times t are computed;
(interpolating line segments only shown to help see)

population increases monotonically to carrying capacity 10
first slowly, then more rapidly, and finally slowing up again

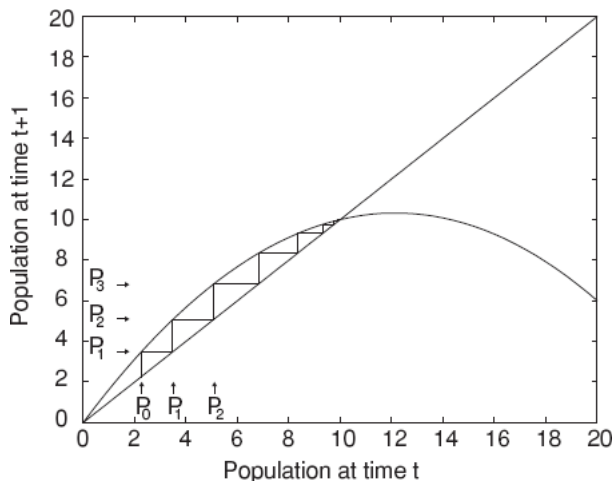
“sigmoidal” picture, as often seen in lab experiments, is observed

useful qualitative tool for understanding nonlinear discrete iterations!
general, but illustrate with previous example

$$P^+ = P(1 + .7(1 - P/10)) , \quad P_0 = 2.3$$

1. graph function that specifies P^+ in terms of P) (parabola, here)
2. graph diagonal line $P^+ = P$
3. mark point $(P_0, P_0) = (2.3, 2.3)$ on the diagonal
4. to find P_1 , move *vertically* toward to the graph of the parabola, to reach a point (P_0, P_1)
5. to find P_2 , first need to mark P_1 on the x-axis, move *horizontally* from (P_0, P_1) toward diagonal, hitting at (P_1, P_1) (kept y-second coordinate same, changed x-coordinate)
6. finally, to find P_2 , move vertically back to parabola to (P_1, P_2)
7. iterate “vertically to parabola, horizontally to diagonal” until pattern obvious

Illustration



clear from graph; whenever initial population $P_0 \in (0, 10)$, population increases, approaching asymptotically carrying capacity K (formally, $\lim_{t \rightarrow \infty} P_t = K$)

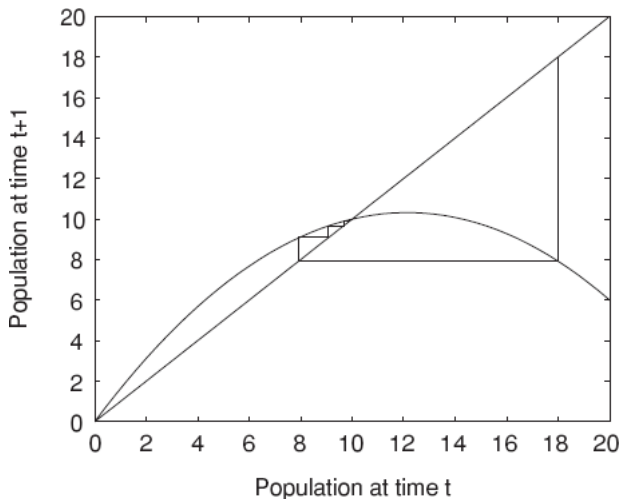
Same example, continued

still with same parameters $r=0.7$ and $K=10$,

if start from $P_0 = 18$ then again $\lim_{t \rightarrow \infty} P_t = K$

approach back to K might be monotonically decreasing, or,

if $P_{t+1} < K$, there is first an “undershoot” followed by a recovery



Starting with higher populations

note: when population starts extremely high,
the population could become negative, which is nonsense
model is still unrealistic
may fix the by truncating parabola at zero (extinction)

Equilibria and linearizations

in logistic example, with parameters $r = 0.7$ and $K = 10$
 $P_t \rightarrow K = 10$ as $t \rightarrow \infty$, no matter what is the initial state,
except if initial population empty, $P_0 = 0$ (stays there)
also if $P_t = 10$ exactly, then $P_{t+1} = 10$
both of $P^* = 0$ and $P^* = 10$ are *steady states*: for $P^+ = F(P)$,

$$F(P^*) = P^*$$

or equivalently, if we write $\Delta P = G(P)$:

$$G(P^*) = 0$$

(other names: *equilibrium* or *fixed point*)

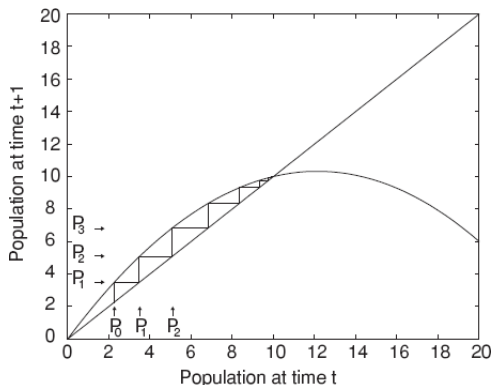
graphically: intersections of graph of $F(P)$ with diagonal

algebraically: solve $P = F(P)$; e.g., solving quadratic

$P = P(1 + .7(1 - P/10))$ for P gives solutions $P = 0$ and $P = 10$

Stability

although both equilibria, major difference between 0 and 10:
population that starts near 0 tends to move away from 0
but a population that starts near 10 tends to move toward 10



$P^* = 0$ is an *unstable* or *repelling* equilibrium
and $P^* = 10$ is a *stable* or *attracting* equilibrium

Definition of stability

mathematically: equilibrium value P^* is said to be (locally, asymptotically) *stable* if this property holds:

for each $\varepsilon > 0$ there is some $\delta > 0$ such that

$$|P_0 - P^*| < \delta \quad \Rightarrow \quad |P_t - P^*| < \varepsilon \quad \forall t > 0 \quad \text{and} \quad \lim_{t \rightarrow \infty} P_t = P^*$$

definition says that provided that we start near P^* , we will never deviate too much from P^* (no “large excursions”), and eventually will asymptotically approach it

but we'll just say “stable” and understand property as saying that $\lim_{t \rightarrow \infty} P_t = P^*$ provided that we start from a P_0 near enough P^*

stability is “local”, only for initial conditions “close enough” to P^* starting from equilibrium $P^*=0$, won't approach stable $P^*=10$

in practice, hard to see unstable equilibria:

smallest perturbation will take us away from that state

Linearization at an equilibrium P^*

look at small deviations from an equilibrium P^* :

$$p_t := P_t - P^* \approx 0$$

$$|p_{t+1}| < |p_t| \Rightarrow P^+ \text{ moves closer to } P^*$$

$$|p_{t+1}| > |p_t| \Rightarrow P^+ \text{ moves away from } P^*$$

so must understand if ratio $\frac{p_{t+1}}{p_t}$ is < 1 or > 1

$$p_{t+1} = P_{t+1} - P^* = F(P_t) - P^* = F(P^* + p_t) - P^*,$$

$$\frac{p_{t+1}}{p_t} = \frac{F(P^* + p_t) - F(P^*)}{p_t} \approx F'(P^*)$$

i.e.

$$p_{t+1} \approx F'(P^*)p_t$$

(recall defn of derivative, $\frac{F(x+h)-F(x)}{h} \rightarrow F'(x)$ as $h \rightarrow 0$)

Stability theorem

Theorem: if P^* is an equilibrium for $P^+ = F(P_t)$, then:

$$\begin{aligned} |F'(P^*)| < 1 &\Rightarrow P^* \text{ stable} \\ |F'(P^*)| > 1 &\Rightarrow P^* \text{ unstable} \end{aligned}$$

$F'(P^*) =$ “linearization” (at the given equilibrium)

e.g. previous example $F(P) = P(1 + .7(1 - P/10))$:

equilibria are 0 and 10

in general, $F'(P) = (1 + .7(1 - P/10)) + P(.7)(-1/10)$,

so $F'(0) = 1.7 > 1$, confirming that 0 is unstable, and

$F'(10) = 1 - .7 = 0.3 < 1$, confirming that 10 is stable

Marginal case

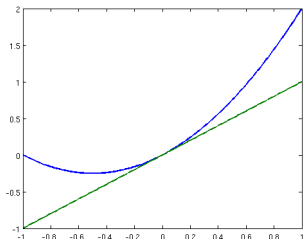
when $|F'(P^*)| = 1$, one cannot decide using just first derivatives

analogy: when checking for local min/max, $f'' = 0$:

min? max? inflection point?

e.g. $F(P) = P + P^2$, $P^* = 0$, $F'(P^*) = 1$:

from $P_0 = -0.01$, convergence to 0; from $P_0 = 0.01$, divergence
so state is neither stable nor unstable



graph of $F(P) = P + P^2$ and diagonal

(optional homework: analyze $F(P) = P + P^3$ and $F(P) = P - P^3$)

Other parameters

still logistic model $P^+ = P(1 + r(1 - P/K))$

for simplicity, $K = 1$ from now on:

$$F(P) = P(1 + r(1 - P)) = P(1 + r - rP)$$

but will investigate behavior for different positive r 's
(before, special case $r = 0.7$)

no matter the value of r , only two equilibria: $P^* = 0$ and $P^* = 1$,

linearizations at each: $F'(0) = 1 + r$, $F'(1) = 1 - r$

so, 0 unstable (for any given value of the parameter $r > 0$)

but what about $P^* = 1$?

now things get really interesting!

we study various examples one at a time

$$0 < r \leq 1$$

since $F'(1) = 1 - r$ and $0 \leq 1 - r < 1$, $P^* = 1$ is stable

moreover: $p_{t+1} \approx F'(1) = (1 - r)p_t$

shows sign of p_t does not change

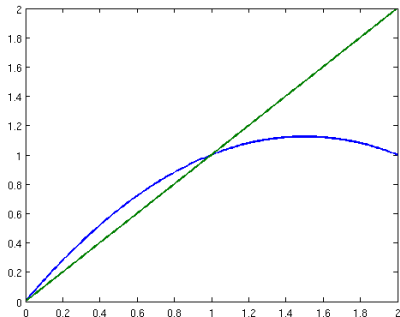
so, not only does the perturbation shrink,

but an initially positive perturbation remains positive

and an initially negative one remains negative

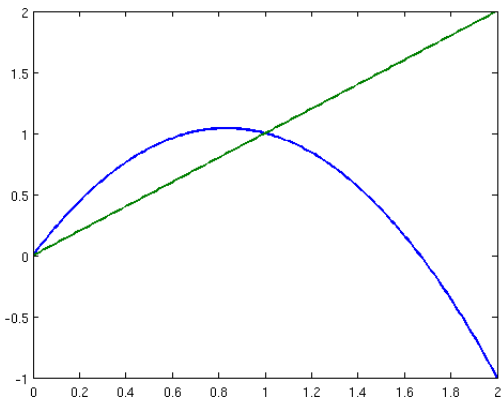
i.e. from $P_0 \approx P^*$, approach equilibrium $P = 1$ without overshoot

verify by cobwebbing:



$$1 < r < 2$$

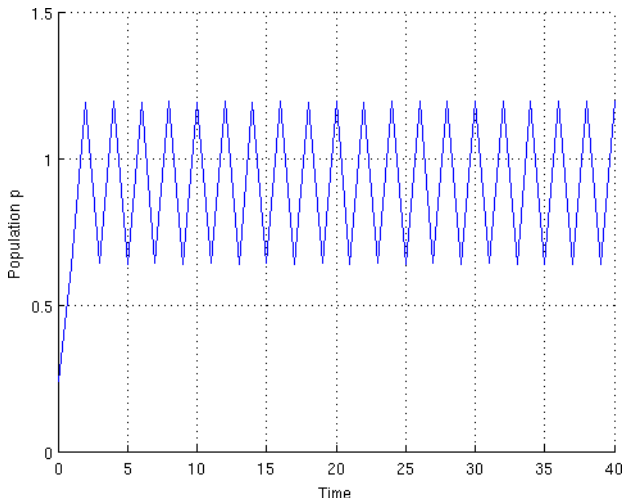
$F'(1) = 1 - r$ and $-1 < 1 - r < 0$, so $P^* = 1$ still stable
 but $p_{t+1} \approx (1 - r)p_t \Rightarrow$ sign of p_t alternates
 expect oscillatory behavior above and below equilibrium
the population approaches equilibrium as a damped oscillation
 see by cobwebbing:



$F(P)$ and diagonal when $r = 0.5$ and $r = 1.5$

$$r > 2$$

$F'(1) = 1 - r$ and $1 - r < -1$, so $P^* = 1$ now unstable
first numerically study special case, $r = 2.4$, plot a few iterates:



iterates when $r = 2.4$, quickly approaching periodic orbit

algebraically find period-2 oscillation, for $r = 2.4$

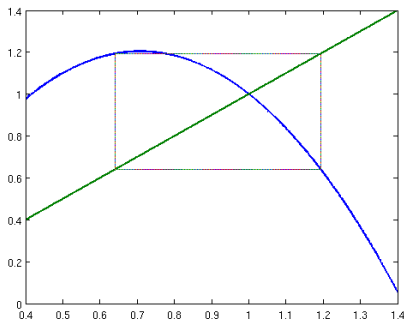
want P_0 such that $P_1 = F(P_0)$ and $P_2 = F(P_1) = P_0$

$$\begin{aligned} F^2(P) &= F(F(P)) = F(P)(3.4 - 2.4F(P)) \\ &= P(3.4 - 2.4P)(3.4 - 2.4P(3.4 - 2.4P)) \end{aligned}$$

solving $F^2(P) = P$: 0 and 1 (they also solve $F(P) = P$)

as well as two new solutions: $\approx 0.6402812675, 1.193052066$

cobwebbing confirms a period-2 oscillation starting from these:

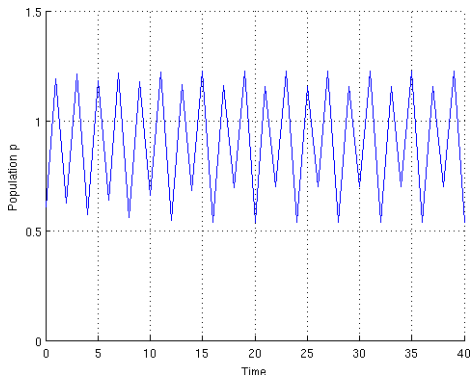


Increasing r

the existence of some 2-cycle persists,...

until at some r_0 a qualitative change occurs, and see a 4-cycle

e.g. $r = 2.5$:

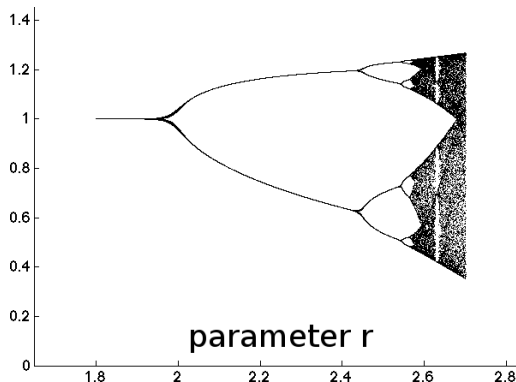


orbit (approx): 1.2250, 0.5359, 1.1577, 0.7012

further increases in r produce an 8-cycle, then a 16-cycle, and so on
example of what is called the “period doubling route to chaos”:

Bifurcations

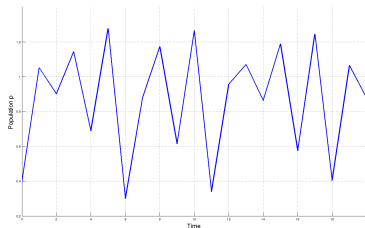
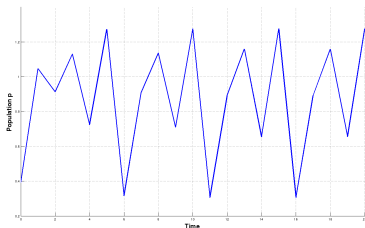
bifurcation diagram: for each value of r , pick some P_0 , then first iterate “enough times” that transient behavior is over; discarding these values, plot future P_t on vertical axis above r :



for $r = 2.4$, two values appear (period-2 orbit we already saw), ...
when r is increased past a certain point ($\approx 2.692 \dots$)
all bifurcations into 2^n -cycles have taken place,
new type of behavior emerges: values seem to be spread out

Chaotic behavior

take for example $r = 2.75$: looks like “random” behavior

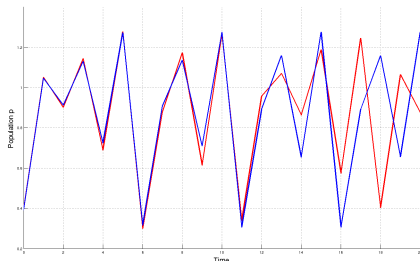


$r = 2.75$, two different initial conditions

but there is nothing random (in the sense of coin-flipping)!
deterministic formula produces the values

sensitive dependence on initial conditions

small change in one state of a deterministic nonlinear system might result in large differences in a later state
populations change similarly only initially, then become very different



“butterfly effect” (mathematician/meteorologist Edward Lorenz, 1961)
metaphor: hurricane influenced by minor perturbations
such as flapping of wings of distant butterfly several weeks earlier
(see also: Jeff Goldblum’s explaining chaos in “Jurassic Park”)

if you are interested in exploring further...

Cushing et al., 1997

chaos in lab population of flour beetle tribolium

